Joint Gaussian Approximation for Group-Sequential Logrank Statistics Beyond Contiguous Alternatives

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Abstract

We study the joint distribution of the logrank statistic evaluated at multiple calendar times in a group-sequential time-to-event trial with staggered entry and administrative censoring. Classical results establish joint asymptotic normality under the null and, via contiguity, under near-null alternatives. We derive a joint Gaussian approximation for the vector of group-sequential logrank statistics under fixed log hazard ratios not restricted to contiguous alternatives. In addition, we provide an exact finite-sample expression for the mean of the logrank statistic at a fixed calendar time. Monte Carlo experiments demonstrate that the proposed centering and covariance approximation substantially improve accuracy relative to null/contiguous approximations for distant alternatives. These results facilitate efficient evaluation of interim and final stopping probabilities and risk-based optimization of group-sequential boundaries.

1 Introduction

In clinical trials where the endpoint of interest is time to failure, the logrank test (Mantel et al., 1966; Peto, 1972) is a popular choice for comparing the survival distributions of the treatments under investigation. Unlike trials which analyze the accumulated data once at termination of the trial, group sequential trials analyze the accumulating data at one or more pre-specified interim time

points over the duration of the trial. Such repeated significance testing using the logrank statistic requires an accurate characterization of its joint distribution computed at the interim time points.

Tsiatis (1981) derived the asymptotic joint distribution of the logrank statistic, computed as the efficient scores test for the proportional hazards model, computed at different time points under the null hypothesis of no treatment difference. This provided a rigorous framework for repeated significance testing in trials with uniform staggered entry, random allocation to different treatments, hazard rate for failure related to the treatment covariate in log-linear fashion, and no loss-to-follow-up censoring. Tsiatis (1982) extended this result to a general class of statistics for testing the hypothesis that the hazard rate for failure is not related to a covariate, in the presence of an independent censoring variable measured from time of entry. The results obtained assuming no difference in survival could be accurately extended to incorporate near-null, contiguous alternatives (Schaid et al., 1990). Applying this approximation with distant non-contiguous alternatives, however, leads to an inaccurately standardized test statistic. Earlier, Hajek (1968) had obtained results on the asymptotic distribution of linear rank statistics in the general regression case without any restrictive assumptions on the score generating function. Since the logrank statistic reduces to a linear rank statistic in the absence of censoring, Hajek (1968) effectively established the asymptotic distribution of the logrank statistic under general alternatives assuming simultaneous entry of all subjects, but restricted to uncensored settings only.

In designing group-sequential trials where the operating characteristics of the design explicitly depend on θ, the treatment effect of interest, accounting for the nature of θ (null, near-null, distant non-null) is crucial to ensuring accurate parameterization of the joint distribution of the group-sequential logrank statistics. Our motivating application is the design of optimal group-sequential trials with time to failure outcomes via minimization of a weighted sum of expected losses where the loss functions incorporate the magnitude of θ. At each interim analysis the logrank statistic is computed using the accumulated data. In the less likely, but clinically plausible, situation of an experimental treatment being far superior to the control/standard treatment ($\theta \ll 0$), computing interim stopping probabilities under the incorrect null asymptotic law may result in an incorrectly standardized test statistic. Consequently, this chapter aims to derive the correct asymptotic distribution theory of the group-sequential logrank statistic under natural assumptions for local or fixed

alternatives with $n^{1/2}\theta_n \to \infty$, including the possibility that $\theta_n = O_P(1)$.

In this chapter, the approximate asymptotic joint distribution of the group-sequentially computed logrank statistics is derived for a class of log hazard ratios not restricted to contiguous alternatives. In the setting of a clinical trial incorporating group-sequential testing at pre-specified times, we assume staggered entry of subjects into the trial followed by random allocation to one of the two treatment arms without the possibility of loss-to-follow-up. At each interim analysis time the logrank statistic is computed utilizing the data available up to that time. We are interested in deriving an approximation to the joint asymptotic distribution of the logrank statistics at the pre-specified interim time points under a broader class of log hazard ratio values not restricted to contiguous alternatives. The proof consists of rewriting the logrank statistic (computed at the calendar time of each analysis) using standard counting process techniques followed by its decomposition into a martingale term and a predictable term (Andersen et al., 2012). The martingale term is expressed as a sum of IID random variables plus a remainder term, that can be shown to be $o_P(1)$ when scaled appropriately, by adjusting the bounded, predictable integrand. Meanwhile, a Taylor expansion in Intermediate-Value form is used to represent the predictable term as the sum total of a deterministic drift term and a sum of IID random variables along with a remainder that can also be shown to be $o_P(1)$ with the proper scaling. This is similar to the approach in Akritas (1984) for establishing the asymptotic normality of a general linear rank statistic for testing the equality of two distributions in the contiguous alternative case. Consequently, the approximate asymptotic multivariate distribution is established using martingale and empirical process limit theorems, with details of the proof along with expressions for the mean, variance, and covariance terms deferred to the Appendix of this chapter.

The remainder of this chapter is organized as follows. Section 2 introduces the required variables followed by the decomposition of the logrank statistic using a standard counting process framework. Section 3 provides the desired approximation to the large sample distribution of the group-sequential logrank statistic computed at pre-specified calendar times, along with an exact analytical expression for its expectation. Results comparing the accuracy of the asymptotic expressions obtained in this chapter with those obtained assuming contiguous alternatives to Monte Carlo simulation based empirical results are provided for a range of log hazard ratios in Section 4. The chapter concludes with

Section 5 discussing the motivation and use of these asymptotic expressions in obtaining analytical approximations to objective functions (weighted sum of expected losses depending explicitly on the magnitude(s) of log hazard ratio(s)) which are to be minimized for deriving optimal group-sequential trials designed to balance time and monetary benefits against the consequences of an incorrectly concluded trial.

2 Notation and Formulas

Consider a clinical trial investigating the effectiveness of a new treatment against a control or a standard treatment. The duration of the clinical trial is pre-specified and fixed at $\tau > 0$. Subjects are assumed to enter the trial at independent random staggered times over the course of the accrual period that ends at t_a ; $0 < t_a < \tau$. The trial has accommodations for repeated significance testing of the accumulated data at pre-specified interim time points through the course of the trial.

Let U be the random variable denoting the calendar time of entry of a subject to the trial. Upon entry, each subject is randomly assigned to one of two treatment arms - standard/control (C) or new (N) - denoted by the random variable Z (0 for C, 1 for N). The time to failure, measured from time of entry, is denoted by the random variable V. Under the Proportional Hazards Model (Cox (1972)), the conditional hazard rate for V is related to the treatment covariate Z as $\lambda_z(t) = \lambda_0(t)e^{\theta z}$. A negative value of the log hazard ratio θ indicates superior effectiveness of the new treatment.

Assume that U is uniformly distributed over the interval $(0, t_a)$. The random treatment indicator Z is distributed as Bernoulli with $P(Z = 1) = p \in (0, 1)$.

Conditional on Z, the failure time V has density $f_{V|Z}(v|z)$, distribution function $F_{V|Z}(v|z) = P(V \leq v|z)$, survival function $\bar{F}_{V|Z}(v|z) = 1 - F_{V|Z}(v|z)$. The conditional hazard rate is $\lambda_z(t) = f_{V|Z}(t|z)/\bar{F}_{V|Z}(t|z)$. The conditional hazard rate given $Z = 0$ is assumed to be known and, by extension, the conditional density, distribution, and survival functions given $Z = 0$ are also known. The corresponding conditional functions given $Z = 1$ can then be determined assuming proportional hazards. Additionally, U is assumed to be independent of (V, Z).

Over the course of the accrual period n subjects are assumed to enter the trial. At time t the available data consists of independently and identically distributed (IID) random vectors $(X_i(t), \Delta_i(t), Z_i)$ for $i = 1, \ldots, n$, where $X_i(t) = \max\{0, \min\{t - U_i, V_i\}\}$ is the time to failure or

(administrative) censoring, and $\Delta_i(t) = I(V_i < (t - U_i)^+)$ is the indicator of failure at time t, for the ith subject. If the time t at which the data has been observed precedes the end of accrual ($t < t_a$), there is a positive probability that not all subjects will have entered the trial. Consequently, the statistic computed at time t will only take into consideration the observed data for subjects who have entered the trial by then.

To test $H_0 : \theta = 0$ at time t, the logrank statistic has the form

$$D(t) = \sum_{i=1}^{n} \Delta_i(t) \left\{ Z_i - \sum_{j \in R(t, X_i(t))} \frac{Z_j}{n(t, X_i(t))} \right\} \tag{1}$$

where $R(t, x)$ is the set of subjects $\{j = 1, \ldots, n\}$ for whom $\{X_j(t) \geq x\}$ and $n(t, x)$ is the number of all such subjects: $n(t, x) = \sum_{i=1}^{n} I(X_i(t) \geq x) = \sum_{i=1}^{n} Y_i(t, x)$, and $Y_i(t, x)$ is the at-risk indicator for the ith subject at time x from entry when observed at calendar time t.

At the calendar time t, define for the ith subject $N_i(t, x) = I(\Delta_i(t) = 1, X_i(t) \leq x)$, a counting process in the time-since-entry argument x, and rewrite the statistic (1) as

$$D(t) = \sum_{i=1}^{n} \int_0^t N_i(t, dx) \left\{ Z_i - \sum_{j \in R(t,x)} \frac{Z_j}{n(t, x)} \right\} = \sum_{i=1}^{n} \int_0^t N_i(t, dx) \left\{ Z_i - \frac{\hat{m}_1(t, x)}{\hat{m}(t, x)} \right\} \tag{2}$$

where $\hat{m}_1(t, x) = n^{-1} \sum_{j=1}^{n} Z_j Y_j(t, x)$ is the average number of subjects receiving treatment N who are at risk at x time units from entry when observed at the calendar time t, and $\hat{m}(t, x) = n^{-1} \sum_{j=1}^{n} Y_j(t, x)$. Under the assumption of proportional hazards, and with respect to the natural filtration $\{\mathcal{F}_{t,x}\}_{0 \leq x \leq t}$ generated by the observed history up to x at calendar time t, the Doob-Meyer decomposition for counting processes gives

$$N_i(t, x) = \Lambda_i(t, x) + M_i(t, x)$$

where $\Lambda_i(t, x) = \int_0^x \lambda_0(s) e^{\theta Z_i} Y_i(t, s)\, ds$ is the $\mathcal{F}_{t,x}$ compensator and $M_i(t, x)$ is an $\mathcal{F}_{t,x}$ martingale. Then by addition and subtraction of the compensator term, we can rewrite (2) as

$$D(t) = \overline{D}(t) + \widetilde{D}(t)$$

where

$$\overline{D}(t) = \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - \lambda_0(x) e^{\theta Z_i} Y_i(t, x)\, dx \Big) \Big\{ Z_i - \frac{\hat{m}_1(t, x)}{\hat{m}(t, x)} \Big\} \tag{3}$$

$$\widetilde{D}(t) = \sum_{i=1}^{n} \int_0^t \lambda_0(x) e^{\theta Z_i} Y_i(t, x) \Big\{ Z_i - \frac{\hat{m}_1(t, x)}{\hat{m}(t, x)} \Big\}\, dx \tag{4}$$

For fixed θ this decomposition of the statistic $D(t)$ into two components: a martingale term $\overline{D}(t)$, and a predictable term $\widetilde{D}(t)$, is a standard starting point to establish asymptotic results for the logrank statistic using martingale theory (Andersen et al. (2012)).

3 Asymptotic Distribution and Exact Expectation of $D(t)$

Theorem 1 provides the approximate asymptotic joint normality of the group-sequential logrank statistics under a fixed, general alternative log hazard ratio θ for a group-sequential clinical trial with $(K-1)$ interim analyses and one final analysis at the pre-specified calendar times $(t_1, \ldots, t_{K-1}, t_K)$: $0 < t_1 < \cdots < t_{K-1} < t_K = \tau$. An exact analytical expression for $\mathbb{E}\widetilde{D}(t)$, and hence $\mathbb{E}D(t)$, at the fixed calendar time t is presented in Theorem 2. The proofs of these theorems are provided in the appendix.

Before we state the two theorems, we first rewrite each of the statistics (3) and (4) as the sum of IID variables plus a remainder. For the martingale term $\overline{D}(\cdot)$ calculated at the calendar time t, we use a first order expansion of $\hat{m}_1(t, x)/\hat{m}(t, x)$ around $m_1(t, x)/m(t, x) = \mathbb{E}[Z_1 Y_1(t, x)]/\mathbb{E}[Y_1(t, x)]$ to obtain:

$$\overline{D}(t) = \overline{D}_A(t) + R_{\overline{D}}(t) = \sum_{i=1}^{n} \psi_{\overline{D},i}(t) + R_{\overline{D}}(t) \tag{5}$$

where

$$\psi_{\overline{D},i}(t) = \int_0^t (N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx)(Z_i - m_1(t, x)/m(t, x)),$$

and $R_{\overline{D}}(t)$ is the remainder such that $n^{-1/2} R_{\overline{D}}(t) = o_P(1)$. For the predictable term $\widetilde{D}(\cdot)$ computed at the calendar time t we first use the risk set averages $\hat{m}_1(t, x)$ and $\hat{m}(t, x)$ to obtain

$$\widetilde{D}(t) = n(e^{\theta} - 1) \int_0^t \lambda_0(x) \hat{m}_1(t, x) \left(1 - \frac{\hat{m}_1(t, x)}{\hat{m}(t, x)} \right) dx = n(e^{\theta} - 1) \int_0^t \lambda_0(x) \frac{\hat{m}_0(t, x) \hat{m}_1(t, x)}{\hat{m}(t, x)}\, dx \tag{6}$$

where $\hat{m}_0(t,x) = \hat{m}(t,x) - \hat{m}_1(t,x) = n^{-1}\sum_{j=1}^{n}(1-Z_j)Y_j(t,x)$. Using a similar approach as before, we derive a second order uniform expansion for $\widetilde{D}(t)$ from (6):

$$\widetilde{D}(t) = \widetilde{D}_A(t) + R_{\widetilde{D}}(t) = n\nu(t) + \sum_{i=1}^{n}\psi_{\widetilde{D},i}(t) + R_{\widetilde{D}}(t) \tag{7}$$

where

$$\begin{aligned}
\nu(t) &= (e^{\theta}-1)\int_0^t \lambda_0(x)\frac{m_0(t,x)m_1(t,x)}{m(t,x)}\,dx \\
\psi_{\widetilde{D},i}(t) &= (e^{\theta}-1)\int_0^t \lambda_0(x)\Bigg[\left(\frac{m_1(t,x)}{m(t,x)}\right)^2\Big((1-Z_i)Y_i(t,x) - m_0(t,x)\Big) \\
&\qquad + \left(\frac{m_0(t,x)}{m(t,x)}\right)^2\Big(Z_iY_i(t,x) - m_1(t,x)\Big)\Bigg]\,dx
\end{aligned}$$

where $m_0(t,x) = m(t,x) - m_1(t,x) = \mathbb{E}[(1-Z_1)Y_1(t,x)]$ and the remainder, as before, satisfies $n^{-1/2}R_{\widetilde{D}}(t) = o_P(1)$. The specifics of the expansions along with proofs of the convergence of the remainder terms are provided in the chapter appendix.

We now state the multivariate central limit theorem we had set out earlier in the section.

Theorem 1. *For K fixed calendar times $0 < t_1 < \cdots < t_{K-1} < t_K \le \tau$, and fixed θ*

$$n^{-1/2}\begin{pmatrix} D(t_1) - n\nu(t_1) \\ \vdots \\ D(t_K) - n\nu(t_K)\end{pmatrix} \xrightarrow{\mathcal{D}} \mathcal{N}_K(\mathbf{0},\Sigma) \qquad \text{as} \qquad n\to\infty \tag{8}$$

where

$$\Sigma_{kk} = \mathbb{V}[\psi_{\overline{D},1}(t_k)] + \mathbb{V}[\psi_{\widetilde{D},1}(t_k)] + 2\mathbb{C}\mathrm{ov}(\psi_{\overline{D},1}(t_k), \psi_{\widetilde{D},1}(t_k)) \tag{9}$$

$$\Sigma_{kl} = \mathbb{C}\mathrm{ov}(\psi_{\overline{D},1}(t_k) + \psi_{\widetilde{D},1}(t_k), \psi_{\overline{D},1}(t_l) + \psi_{\widetilde{D},1}(t_l)) \tag{10}$$

for $k,l = 1,\ldots,K$, and $k \ne l$.

Theorem 2. *At the calendar time $t\,(0 < t < \tau)$, and for fixed θ*

$$\mathbb{E}[\widetilde{D}(t)] = n(e^{\theta}-1)\int_0^t \lambda_0(x)\frac{m_0(t,x)m_1(t,x)}{m(t,x)}\left(1 - \frac{1-(1-\frac{m_1(t,x)}{p})^n}{n\frac{m_1(t,x)}{p}}\right)dx \tag{11}$$

Substituting the exact expectation (11) in place of $n\nu(\cdot)$ in (8) retains the asymptotic normality because $n^{-1/2}(\mathbb{E}\widetilde{D}(t) - n\nu(t)) = o(1)$.

4 Simulation Study

The objective of this simulation study is to assess and compare the finite-sample accuracy of the existing expressions for asymptotic means, variances, and covariances of the group-sequentially computed logrank statistic under null or contiguous alternative log hazard ratios (Tsiatis, 1981, 1982; Schaid et al., 1990) against those proposed in this paper for a general class of alternatives.

We consider a two-arm clinical trial of fixed duration $\tau = 3$ years with subject accrual lasting $t_a = 2$ years. The accumulating data is analyzed at three interim time points and once at the end of the trial: $(t_1, t_2, t_3, t_4) = (1.5, 2, 2.5, 3)$ years. We assume a fixed number of subjects enter the trial in a staggered, uniform manner over the course of the accrual period, with no loss-to-follow-up and that administrative censoring occurs at $\tau = 3$ years.

We consider $\theta \in \{-1.5, -1.4, \ldots, -0.1\}$, and $n = 300$ patients for the purpose of this simulation study. For each θ, we generate $B = 100{,}000$ independent replicates of the clinical trial:

$$\mathbf{U}_b = (U_{b1}, \ldots, U_{bn}) \ : \ U_{b1}, \ldots, U_{bn} \overset{IID}{\sim} \text{Uniform}(0, t_a = 2)$$
$$\mathbf{Z}_b = (Z_{b1}, \ldots, Z_{bn}) \ : \ Z_{b1}, \ldots, Z_{bn} \overset{IID}{\sim} \text{Bernoulli}(p = 0.5)$$
$$\mathbf{V}_b^{(0)} = (V_{b1}^{(0)}, \ldots, V_{bn}^{(0)}) \ : \ V_{b1}^{(0)}, \ldots, V_{bn}^{(0)} \overset{IID}{\sim} \text{Exponential}(\text{rate} = \lambda_0 = 1)$$

and obtain the vector of failure times by transforming the vector $\mathbf{V}_b^{(0)}$ as follows:

$$\left[\mathbf{V}_b = (e^{-\theta Z_{b1}} V_{b1}^{(0)}, \ldots, e^{-\theta Z_{bn}} V_{bn}^{(0)})\right] \qquad \text{for} \qquad b = 1, \ldots, B$$

so that $(V_{bi}|Z_{bi} = z) \sim \text{Exponential}(\text{rate} = \lambda_z)$ for all $i = 1, \ldots, n$. Using $(\mathbf{U}_b, \mathbf{V}_b, \mathbf{Z}_b)$ in (1), we compute the bth replicate of the vector of logrank statistics at the calendar times $(t_1, \ldots, t_4)$:

$$\mathbf{D}_b = (D_b(t_1), \ldots, D_b(t_4)) \qquad \text{for} \qquad b = 1, \ldots, B$$

The mean vector of $n^{-1/2}(D(t_1), \ldots, D(t_4))$ is estimated using the sample average

$$\widehat{\mu} = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{\sqrt{n}} \mathbf{D}_b \tag{12}$$

with component-wise estimated standard error

$$SE(\widehat{\mu}_k) = \frac{1}{\sqrt{B}} \left[\frac{1}{B-1} \sum_{b=1}^{B} \left(\frac{D_b(t_k)}{\sqrt{n}} - \widehat{\mu}_k \right)^2 \right]^{1/2} \qquad \text{for} \qquad k = 1, \ldots, 4 \tag{13}$$

The asymptotic covariance matrix Σ is estimated by the sample covariance matrix

$$\widehat{\Sigma} = \frac{1}{B-1} \sum_{b=1}^{B} \left(\frac{\mathbf{D}_b}{\sqrt{n}} - \widehat{\mu} \right) \left(\frac{\mathbf{D}_b}{\sqrt{n}} - \widehat{\mu} \right)^T$$

with componentwise estimated standard error

$$SE(\widehat{\Sigma}_{kl}) = \left[\frac{1}{B-1} \left(\widehat{\Sigma}_{kk} \widehat{\Sigma}_{ll} + \widehat{\Sigma}_{kl}^2 \right) \right]^{1/2} \qquad \text{for} \qquad 1 \leq k, l \leq 4$$

We compare $\widehat{\mu}(\theta)$ and $\widehat{\Sigma}(\theta)$ to the corresponding theoretical quantities from (i) Tsiatis (1981) and Schaid et al. (1990) assuming null or contiguous alternatives and (ii) Section 3 for a general class of alternatives. For (i) we use

$$\Sigma_{kl}^C = p\,(1-p)\,P(\Delta_1(t_k \wedge t_l) = 1) \qquad \text{for} \quad 1 \leq k, l \leq 4 \tag{14}$$

$$\mu_k^C = \sqrt{n}\,\theta \sqrt{\Sigma_{kk}^C} \qquad \text{for} \qquad 1 \leq k \leq 4 \tag{15}$$

obtained from Theorem 3.1 in Tsiatis (1981). For (ii) we use the exact expression (11) for the mean,

$$\mu^G = n^{-1/2}(\mathbb{E}D(t_1), \ldots, \mathbb{E}D(t_4)) = n^{-1/2}(\mathbb{E}\,\widetilde{D}(t_1), \ldots, \mathbb{E}\,\widetilde{D}(t_4))$$

and the expressions (9) and (10) for the variances and covariances, respectively, with the corresponding covariance matrix denoted by Σ^G.

The results are summarized in the following tables. For the sake of brevity we only tabulate the complete results for $\theta \in \{-1.5, -0.1\}$. We report the mean vectors in their entirety along with the

component-wise estimated standard errors for $\widehat{\mu}$. To highlight the accuracy of the exact expectation in (11) over the approximation μ^C (assuming contiguous alternatives), we report the ℓ_∞ norm of each of μ^G and μ^C relative to the estimated mean $\widehat{\mu}$:

$$\|\hat{\mu} - \mu^M\|_\infty = \max_{1 \le k \le 4} |\hat{\mu}_k - \mu_k^M| \tag{16}$$

We also represent this ℓ_∞ norm comparison visually for all $\theta \in \{-1.5, -1.4, \ldots, -0.1\}$ to showcase the rapidly worsening accuracy of μ^C as compared to that of μ^G as θ decreases. For comparing the covariance matrices Σ^G and Σ^C, we report the maximum norm (maximum absolute entry-wise difference) of each of them relative to $\widehat{\Sigma}$ as a primary metric of discrepancy, and the Frobenius norm as a summary of overall error

$$\|\widehat{\Sigma} - \Sigma^M\|_{\max} = \max_{1 \le k,l \le 4} \left|\widehat{\Sigma}_{kl} - \Sigma_{kl}^M\right| \tag{17}$$

$$\|\widehat{\Sigma} - \Sigma^M\|_F = \left[\sum_{k=1}^{4}\sum_{l=1}^{4}\left(\widehat{\Sigma}_{kl} - \Sigma_{kl}^M\right)^2\right]^{1/2} \qquad \text{for} \qquad M \in \{C, G\} \tag{18}$$

θ		
-1.5	μ^C	$(-1.5399, -2.4714, -3.3004, -3.8829)$
	μ^G	$(-1.3832, -2.2353, -2.9934, -3.5074)$
	$\hat{mu}$	$(-1.3844, -2.2372, -2.9970, -3.5113)$
	$SE(\hat{\mu})$	$(0.0007, 0.0009, 0.001, 0.001)$
-0.1	μ^C	$(-0.1518, -0.2394, -0.3126, -0.3581)$
	μ^G	$(-0.1492, -0.2367, -0.3092, -0.3540)$
	$\hat{mu}$	$(-0.1502, -0.2361, -0.3098, -0.3551)$
	$SE(\hat{\mu})$	$(0.0009, 0.0012, 0.0013, 0.0014)$

Table 1: Comparison of the mean approximation (15) and the exact mean (11) to the estimated sample mean (12). The component-wise standard errors (13) for $\widehat{\mu}$ are also provided.

θ	$M = C/G$	$\|\hat{\mu} - \mu^M\|_\infty$	$\|\widehat{\Sigma} - \Sigma^M\|_{\max}$	$\|\widehat{\Sigma} - \Sigma^M\|_F$
-1.5	C	0.3716	0.0560	0.1222
	G	0.0039	0.0037	0.0074
-0.1	C	0.0033	0.0032	0.007
	G	0.0011	0.0020	0.005

Table 2: Comparison of the ℓ_∞ norm for the mean approximations, along with the max norm and the Frobenius norm for the variance/covariance approximations.

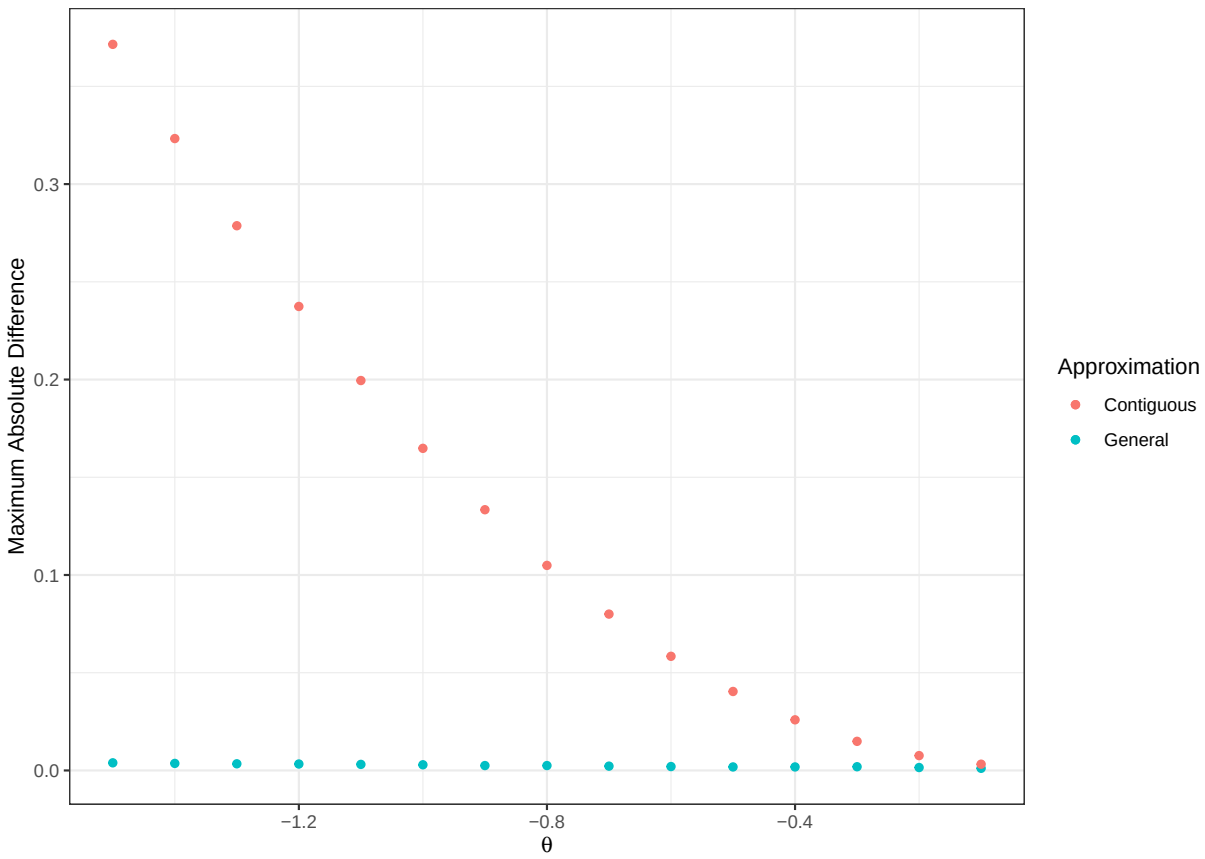

Figure 1: $\|\widehat{\mu} - \mu^M\|_\infty$; $M \in \{C, G\}$ computed for $\theta \in \{-1.5, -1, 4, \ldots, -0.1\}$.

Across all values of θ considered, the general θ approximations from Section 3 track the Monte Carlo means and second moments substantially more closely than the null/contiguous approximations. The discrepancy under the null/contiguous approach increases as the effect size grows: for $\theta = -1.5$ the maximum absolute deviation in the mean is 0.3716 under μ^C versus 0.0039 under μ^G, as shown in Table 2, with analogous improvements for the covariance matrix. The larger improvement in the mean is consistent with the fact that μ^G uses the exact finite-sample expression for $\mathbb{E}[n^{-1/2}D(t_k)]$, whereas μ^C relies on a local (contiguous) approximation. Furthermore, this discrepancy in accuracy between the two mean approximations increases rapidly as θ becomes more negative as evidenced in Figure 1.

5 Discussion

This paper extends the work of Tsiatis (1981) by deriving a multivariate normal approximation to the joint distribution of the group-sequential logrank statistics computed at fixed calendar times

under fixed (potentially non-contiguous) log hazard ratios, rather than restricting attention to the null or contiguous alternatives. In addition, it provides an exact, finite-sample expression of the expectation of the logrank statistic. These results complement Schaid et al. (1990) by providing the exact centering term and covariance structure for effect sizes not local to the null.

The primary motivation behind this work is the construction of group-sequential boundaries in trials with time-to-failure outcomes by minimizing an objective function that is the weighted sum of expected loss that depend explicitly on the magnitude of the log hazard ratio. The objective function averages over multiple clinically plausible values of the effect size and depends on interim stopping probabilities. The results of this paper provide a means to accurately compute interim stopping probabilities involving the group-sequential logrank statistics as multivariate Gaussian probabilities using an appropriate centering and covariance approximation. This can be directly incorporated into the evaluation of the objective function to be optimized to derive optimal boundaries.

The results are derived assuming proportional hazards, uniform staggered entry over a fixed accrual period, and administrative censoring at termination of the trial with no loss-to-follow-up. They can be extended to incorporate an independent censoring mechanism as in Tsiatis (1982) using the same martingale decomposition followed by an asymptotic linear expansion strategy. However, the resulting mean and variance/covariance expressions may become more cumbersome. The same framework can also be adapted for the case of the weighted-logrank statistic computed over time, as well as non-proportional hazards.

Appendix

To facilitate the discussion of the proof of the multivariate CLT in **Theorem 1** and obtain the exact finite-sample expectation in **Theorem 2**, we reintroduce the required statistics. The logrank statistic at time $t \in [0, \tau]$, denoted by $D(t)$, is expressed as:

$$D(t) = \sum_{i=1}^{n} \Delta_i(t) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, X_i(t))}{\sum_{j=1}^{n} Y_j(t, X_i(t))} \right\} = \sum_{i=1}^{n} \int_0^t N_i(t, dx) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, x)}{\sum_{j=1}^{n} Y_j(t, x)} \right\}$$

where $Y_i(t, x) = I(X_i(t) \geq x)$, and $N_i(t, x) = I(\Delta_i(t) = 1, X_i(t) \leq x)$ for $x \in [0, t]$. We can decompose $D(t)$ into two components by adding and subtracting the appropriate compensator from the counting process:

$$\begin{aligned}
D(t) &= \overline{D}(t) + \widetilde{D}(t) \\
\overline{D}(t) &= \sum_{i=1}^{n} \int_0^t \left(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx \right) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, x)}{\sum_{j=1}^{n} Y_j(t, x)} \right\} \\
\widetilde{D}(t) &= \sum_{i=1}^{n} \int_0^t e^{\theta Z_i} \lambda_0(x) Y_i(t, x) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, x)}{\sum_{j=1}^{n} Y_j(t, x)} \right\} dx
\end{aligned}$$

where $e^{\theta Z} = Z e^{\theta} + (1 - Z)$. For $t \in [0, \tau]$ and $x \in [0, t]$, we define the treatment-wise risk set averages

$$\hat{m}_1(t, x) = n^{-1} \sum_{j=1}^{n} Z_j Y_j(t, x) \tag{19}$$

$$\hat{m}_0(t, x) = n^{-1} \sum_{j=1}^{n} (1 - Z_j) Y_j(t, x) \tag{20}$$

and the combined risk set average

$$\hat{m}(t, x) = \hat{m}_1(t, x) + \hat{m}_0(t, x) = n^{-1} \sum_{j=1}^{n} Y_j(t, x) \tag{21}$$

We re-express $\overline{D}(t)$ and $\widetilde{D}(t)$ in terms of these risk set averages:

$$\overline{D}(t) = \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx \Big) \left\{ Z_i - \frac{\hat{m}_1(t, x)}{\hat{m}(t, x)} \right\} \tag{22}$$

$$\widetilde{D}(t) = n\,(e^{\theta} - 1) \int_0^t \lambda_0(x)\, \frac{\hat{m}_0(t, x)\, \hat{m}_1(t, x)}{\hat{m}(t, x)}\, dx \tag{23}$$

Corresponding to the treatment-wise risk set averages $\hat{m}_z(t, x)$; $z \in \{0, 1\}$, we define their asymptotic counterparts

$$m_1(t, x) = \mathbb{E}[Z_1 Y_1(t, x)] = p\, P(X_1(t) \geq x | Z_1 = 1) \tag{24}$$

$$m_0(t, x) = \mathbb{E}[(1 - Z_1) Y_1(t, x)] = (1 - p)\, P(X_1(t) \geq x | Z_1 = 0) \tag{25}$$

and

$$m(t, x) = m_0(t, x) + m_1(t, x) = \mathbb{E}[Y_1(t, x)] = P(X_1(t) \geq x) \tag{26}$$

corresponding to the combined average $\hat{m}(t, x)$.

In the rest of this Appendix, we assume that the subject-level observations (U_i, V_i, Z_i) are independent for $i = 1, \ldots, n$. In addition, we assume that the cumulative hazard on the control arm $(Z = 0)$ at the scheduled end of the trial τ is finite: $\Lambda_0(\tau) = \int_0^\tau \lambda_0(x) dx < \infty$.

In order to prove the required multivariate CLT, we first establish the following Glivenko-Cantelli type lemmas for the risk set averages.

Lemma 1. *For $t \in [0, \tau]$*

$$\sup_{0 \leq x \leq t} |\hat{m}_1(t, x) - m_1(t, x)| = O_P(n^{-1/2}) \tag{27}$$

Proof. For $t \in [0, \tau]$, we define the stochastic process

$$\xi(x) = Z\, Y(t, x) = Z\, I(X(t) \geq x), \qquad x \in [0, t]$$

where $X(t) = \min\{(t - U)^+, V\}$, and (U, V, Z) has the same distribution as (U_1, V_1, Z_1). For the

empirical measure $\mathbb{P}_n$, we can write

$$\hat{m}_1(t,x) = \mathbb{P}_n \xi(x) \qquad \text{and} \qquad m_1(t,x) = P\xi(x)$$

such that

$$n^{1/2}(\hat{m}_1(t,x) - m_1(t,x)) = n^{1/2}(\mathbb{P}_n - P)\xi(x)$$

Since $Z \in \{0,1\}$ and $Y(t,x) \in \{0,1\}$ for all $x \in [0,t]$ is non-increasing, and right-continuous with left-hand limits in $x \in [0,t]$, $\{\xi(x) : x \in [0,t]\}$ is a bounded, monotone, càdlàg stochastic process. Furthermore, since $|\xi(x)| \leq 1$ for $x \in [0,t]$,

$$P[\,(|\xi(0)| \vee |\xi(t)|)^2\,] \leq 1 < \infty$$

By (Kosorok, 2008, Lemma 4.1) ξ is P-Donsker. Therefore, for $t \in [0,\tau]$, the class of functions

$$\mathbb{F}_t = \{f_x : (u,v,z) \mapsto z\, I(\min\{(t-u)^+, v\} \geq x)\,;\, 0 \leq x \leq t\}$$

is P-Donsker. Consequently, the empirical process

$$G_n(x) = n^{1/2}(\mathbb{P}_n - P)f_x = n^{1/2}(\hat{m}_1(t,x) - m_1(t,x)) \qquad x \in [0,t]$$

converges weakly in $\ell^\infty([0,t])$ to a tight, mean-zero Gaussian process. By the continuous mapping theorem with

$$g : \ell^\infty[0,t] \to \mathbb{R}, \qquad g(h) = \|h\|_\infty = \sup_{0 \leq x \leq t} |h(x)|\ ,$$

$$\|G_n\|_\infty = \sup_{0 \leq x \leq t} |n^{1/2}(\hat{m}_1(t,x) - m_1(t,x))| \xrightarrow{\mathcal{D}} \|G\|_\infty$$

where G is the Gaussian limit. Since $\|G\|_\infty < \infty$ almost surely, it follows that

$$\sup_{0 \leq x \leq t} |(\hat{m}_1(t,x) - m_1(t,x))| = O_P(n^{-1/2})$$

□

Corollary 1. *For* $t \in [0, \tau]$

$$\sup_{0 \le x \le t} |\hat{m}_0(t,x) - m_0(t,x)| = O_P(n^{-1/2}) \tag{28}$$

$$\sup_{0 \le x \le t} |\hat{m}(t,x) - m(t,x)| = O_P(n^{-1/2}) \tag{29}$$

We can now begin to prove the multivariate CLT in **Theorem 1**.

Proof of Theorem 1

The proof has the following three main parts:

1. Express $\overline{D}(t)$ and $\widetilde{D}(t)$ for $t \in [0, \tau]$ as the sum of IID terms plus an $o_P(n^{1/2})$ remainder.

2. Establish the multivariate CLT for the scaled and centered logrank statistics.

3. Obtain the expressions for the asymptotic variances and covariances.

Part 1: Decompose $\overline{D}(t)$ and $\widetilde{D}(t)$ as a sum of IID subject-level contributions plus a remainder

At time $t \in [0, \tau]$, by adding and subtracting $m_1(t,x)/m(t,x)$ in (22), we can rewrite $\overline{D}(t)$ as:

$$\overline{D}(t) = \overline{D}_A(t) + R_{\overline{D}}(t) \tag{30}$$

$$\overline{D}_A(t) = \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t,x)\, dx \Big) \left(Z_i - \frac{m_1(t,x)}{m(t,x)} \right) \tag{31}$$

$$R_{\overline{D}}(t) = \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t,x)\, dx \Big) \left(\frac{m_1(t,x)}{m(t,x)} - \frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} \right) \tag{32}$$

where $\overline{D}_A(t)$ is a sum of IID terms, and we shall prove $n^{-1/2}R_{\overline{D}}(t) = o_P(1)$. Since $R_{\overline{D}}(t)$ is the sum of n orthogonal, zero-mean, square integrable martingales we can write

$$\begin{aligned}\mathbb{V}(n^{-1/2}R_{\overline{D}}(t)) &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{V}\left[\int_0^t \Big(N_i(t,dx) - e^{\theta Z_i}\lambda_0(x)Y_i(t,x)\,dx\Big)\left(\frac{m_1(t,x)}{m(t,x)} - \frac{\hat{m}_1(t,x)}{\hat{m}(t,x)}\right)\right] \\ &= \mathbb{E}\left[\int_0^t \left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t,x)dx\right] \qquad (33)\end{aligned}$$

where the second equality is established by noting that variance of each martingale term is the expectation of its predictable quadratic variation.

We examine the behavior of the integrand in (33) depending on the size of the risk set average $\hat{m}(t,x)$, and the size of its asymptotic counterpart $m(t,x)$. On the set

$$A_n(t) = \{x \in [0,t] : \hat{m}(t,x), m(t,x) > \delta_t\} \qquad \text{for some} \qquad \delta_t > 0 \qquad (34)$$

we have

$$\begin{aligned}\left|\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right| &\le \left|\frac{1}{\hat{m}(t,x)}\right||\hat{m}_1(t,x) - m_1(t,x)| + \left|\frac{m_1(t,x)}{m(t,x)}\right|\left|\frac{1}{\hat{m}(t,x)}\right||\hat{m}(t,x) - m(t,x)| \\ &\le \frac{1}{\delta_t}\sup_{0\le x\le t}|\hat{m}_1(t,x) - m_1(t,x)| + \frac{1}{\delta_t}\sup_{0\le x\le t}|\hat{m}(t,x) - m(t,x)|\end{aligned}$$

using $m_1(t,x)/m(t,x) \le 1$. Since $(a+b)^2 \le 2(a^2+b^2)$, we can write

$$\left[\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right]^2 \le \frac{2}{\delta_t^2}\left[\Big(\sup_{0\le x\le t}|\hat{m}_1(t,x) - m_1(t,x)|\Big)^2 + \Big(\sup_{0\le x\le t}|\hat{m}(t,x) - m(t,x)|\Big)^2\right]$$

Using this in (33), we can write

$$\begin{aligned}&\mathbb{E}\left[\int_{[0,t]\cap A_n(t)}\left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t,x)dx\right] \\ &\le \frac{2(1+e^\theta)}{\delta_t^2}\mathbb{E}\left[\int_{[0,t]\cap A_n(t)}\lambda_0(x)\Big(\sup_{0\le x\le t}|\hat{m}_1(t,x) - m_1(t,x)|\Big)^2 + \Big(\sup_{0\le x\le t}|\hat{m}(t,x) - m(t,x)|\Big)^2 dx\right] \\ &\le \frac{2(1+e^\theta)}{\delta_t^2}\,\Lambda_0(\tau)\,\mathbb{E}\left[\Big(\sup_{0\le x\le t}|\hat{m}_1(t,x) - m_1(t,x)|\Big)^2 + \Big(\sup_{0\le x\le t}|\hat{m}(t,x) - m(t,x)|\Big)^2\right]\end{aligned}$$

using $e^{\theta Z_i} \leq (1+e^{\theta})$ and $Y_i(t,x) \in \{0,1\}$ for all $x \in [0,t]$. Then using the results (27) and (29), since $0 \leq |\hat{m}_1(t,x) - m_1(t,x)|, |\hat{m}(t,x) - m(t,x)| \leq 1$, we can conclude

$$\mathbb{E}\left[\Big(\sup_{0\leq x\leq t} |\hat{m}_1(t,x) - m_1(t,x)|\Big)^2\right] \to 0\ ,\ \mathbb{E}\left[\Big(\sup_{0\leq x\leq t} |\hat{m}(t,x) - m(t,x)|\Big)^2\right] \to 0 \tag{35}$$

Therefore,

$$\mathbb{E}\left[\int_{[0,t]\cap A_n(t)} \left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t,x)dx\right] \to 0$$

Instead of directly considering the set $A_n^c(t)$, we consider the two sets

$$B_{n1}(t) = \{x \in [0,t] : m(t,x) < 2\delta_t\},\ \text{and} \tag{36}$$

$$B_{n2}(t) = \{x \in [0,t] : \hat{m}(t,x) < \delta_t, m(t,x) \geq 2\delta_t\} \tag{37}$$

For the set $B_{n2}(t)$:

$$\begin{aligned} P(B_{n2}(t)) &= P\big(\{x \in [0,t] : \hat{m}(t,x) < \delta_t, m(t,x) \geq 2\delta_t\}\big) \\ &\leq P\big(\{x \in [0,t] : |\hat{m}(t,x) - m(t,x)| > \delta_t\}\big) \to 0 \end{aligned} \tag{38}$$

using (29). Therefore,

$$\begin{aligned} \mathbb{E}\left[\int_{[0,t]\cap B_{n2}(t)} \left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t,x)dx\right] &\leq (1+e^{\theta})\mathbb{E}\left[\int_0^t \lambda_0(x)\, I(B_{n2}(t))\, dx\right] \\ &\leq (1+e^{\theta})\Lambda_0(\tau)P(B_{n2}(t)) \to 0 \end{aligned} \tag{39}$$

using

$$\left[\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right]^2 \leq 1\,, \qquad e^{\theta Z_i} \leq (1+e^{\theta})\,, \qquad Y_i(t,x) \in \{0,1\}$$

On the set $B_{n1}(t)$:

$$
\begin{aligned}
&\mathbb{E}\left[\int_{[0,t]\cap B_{n1}(t)} \left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t,x)dx\right] \\
&\leq (1+e^{\theta})\mathbb{E}\left[\int_{[0,t]\cap B_{n1}(t)} \lambda_0(x)Y_1(t,x)dx\right] \\
&= (1+e^{\theta})\int_0^t \lambda_0(x)\,\mathbb{E}[Y_1(t,x)\,I(B_{n1}(t))]\,dx \\
&\leq (1+e^{\theta})\,\Lambda_0(\tau)\,2\delta_t
\end{aligned}
$$

using $\left[\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right]^2 \leq 1$. Noting that $\delta_t > 0$ can be chosen to be arbitrarily small, we have

$$
\mathbb{E}\left[\int_{[0,t]\cap B_{n1}(t)} \left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t,x)dx\right] \to 0 \tag{40}
$$

Combining (35) , (39) and (40), we have

$$
\mathbb{V}[n^{-1/2}R_{\overline{D}}(t)] \to 0 \implies n^{-1/2}R_{\overline{D}}(t) = o_P(1) \tag{41}
$$

Therefore, for $t \in [0,\tau]$

$$
n^{-1/2}\overline{D}(t) = n^{-1/2}\overline{D}_A(t) + o_P(1)
$$

We treat $\widetilde{D}(t)$ in (23) similarly: for $t \in [0,\tau]$:

$$
\begin{aligned}
\widetilde{D}(t) &= n\,(e^{\theta}-1)\int_0^t \lambda_0(x)\,\frac{\hat{m}_0(t,x)\,\hat{m}_1(t,x)}{\hat{m}(t,x)}\,dx \\
&= n\,(e^{\theta}-1)\int_0^t \lambda_0(x)\,g\Big(\hat{m}_0(t,x)\,,\,\hat{m}_1(t,x)\Big)\,dx
\end{aligned}
$$

where $g(a,b) = ab/(a+b)$. Using a first order Taylor series in Intermediate-Value form for $g\big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x)\big)$

where $\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) = \big(\hat{m}_0(t,x)\,,\,\hat{m}_1(t,x)\big)$ around $\underset{\sim}{\mathbf{m}}_{01}(t,x) = \big(m_0(t,x)\,,\,m_1(t,x)\big)$, we can write:

$$\begin{aligned} g(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x)) - g(\underset{\sim}{\mathbf{m}}_{01}(t,x)) - \nabla^T g(\underset{\sim}{\mathbf{m}}_{01}(t,x))\big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) - \underset{\sim}{\mathbf{m}}_{01}(t,x)\big) \\ = \big(\nabla^T g(\overline{\underset{\sim}{\mathbf{m}}}_{01}(t,x)) - \nabla^T g(\underset{\sim}{\mathbf{m}}_{01}(t,x))\big)\big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) - \underset{\sim}{\mathbf{m}}_{01}(t,x)\big) \end{aligned}$$

where $\nabla g(\underset{\sim}{\mathbf{m}})$ is the gradient of g computed at $\underset{\sim}{\mathbf{m}}$, and $\overline{\underset{\sim}{\mathbf{m}}}_{01}(t,x)$ lies on the line segment joining $\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x)$ and $\underset{\sim}{\mathbf{m}}_{01}(t,x)$. We can then rewrite $\widetilde{D}(t)$ for $t \in [0,\tau]$ as:

$$\widetilde{D}(t) = \widetilde{D}_A(t) + R_{\widetilde{D}}(t) \tag{42}$$

$$\widetilde{D}_A(t) = n(e^\theta - 1)\int_0^t \lambda_0(x)\Big[g(\underset{\sim}{\mathbf{m}}_{01}(t,x)) + \nabla^T g(\underset{\sim}{\mathbf{m}}_{01}(t,x))\big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) - \underset{\sim}{\mathbf{m}}_{01}(t,x)\big)\Big]dx \tag{43}$$

$$R_{\widetilde{D}}(t) = n(e^\theta - 1)\int_0^t \lambda_0(x)\Big[\big(\nabla^T g(\overline{\underset{\sim}{\mathbf{m}}}_{01}(t,x)) - \nabla^T g(\underset{\sim}{\mathbf{m}}_{01}(t,x))\big)\big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) - \underset{\sim}{\mathbf{m}}_{01}(t,x)\big)\Big]dx \tag{44}$$

As before, we shall show $n^{-1/2}R_{\widetilde{D}}(t) = o_P(1)$ by considering its behavior over the same three sets $A_n(t)$, $B_{n1}(t)$, and $B_{n2}(t)$.

$$n^{-1/2}R_{\widetilde{D}}(t) = (e^\theta - 1)\int_0^t \lambda_0(x)\Big[\big(\nabla^T g(\overline{\underset{\sim}{\mathbf{m}}}_{01}(t,x)) - \nabla^T g(\underset{\sim}{\mathbf{m}}_{01}(t,x))\big)\, n^{1/2}\,\big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) - \underset{\sim}{\mathbf{m}}_{01}(t,x)\big)\Big]dx$$

Noting that

$$\nabla g(\overline{\underset{\sim}{\mathbf{m}}}_{01}(t,x)) - \nabla g(\underset{\sim}{\mathbf{m}}_{01}(t,x)) = \left[\left(\frac{\overline{m}_1(t,x)}{\overline{m}(t,x)}\right)^2 - \left(\frac{m_1(t,x)}{m(t,x)}\right)^2, \left(\frac{\overline{m}_0(t,x)}{\overline{m}(t,x)}\right)^2 - \left(\frac{m_0(t,x)}{m(t,x)}\right)^2\right]^T$$

where $\overline{m}(t,x) = \overline{m}_0(t,x) + \overline{m}_1(t,x)$, on the set $A_n(t) = \{x \in [0,t] : \hat{m}(t,x), m(t,x) > \delta_t\}$ we can use

$$\left|\left(\frac{\overline{m}_z(t,x)}{\overline{m}(t,x)}\right)^2 - \left(\frac{m_z(t,x)}{m(t,x)}\right)^2\right| = o_P(1), \qquad \text{and}$$

$$n^{1/2}\big(\hat{m}_z(t,x) - m_z(t,x)\big) \le n^{1/2}\sup_{0\le x\le t}\big|\hat{m}_z(t,x) - m_z(t,x)\big| = O_P(1) \qquad \text{for} \qquad z \in \{0,1\}$$

to show that by restricting the integral in $n^{-1/2}R_{\widetilde{D}}(t)$ to $[0,t] \cap A_n(t)$

$$|n^{-1/2}R_{\widetilde{D}}(t)| \xrightarrow{P} 0$$

For the set $B_{n2}(t) = \{x \in [0,t] : \hat{m}(t,x) < \delta_t, m(t,x) \geq 2\delta_t\}$, as in (38), here too we have

$$P\big(B_{n2}(t)\big) \to 0$$

We can treat $n^{-1/2}R_{\widetilde{D}}(t)$ on the set $B_{n1}(t)$ in a similar manner as before in (40), using

$$\left| \left(\frac{\overline{m}_z(t,x)}{\overline{m}(t,x)} \right)^2 - \left(\frac{m_z(t,x)}{m(t,x)} \right)^2 \right| \leq 1 \qquad \text{for} \qquad z \in \{0,1\}$$

to show that on $B_{n1}(t)$:

$$\begin{aligned} |n^{-1/2}R_{\widetilde{D}}(t)| \leq |e^{\theta} - 1| & \\ & \sqrt{n}\Big(\sup_{0 \leq x \leq t} |\hat{m}_1(t,x) - m_1(t,x)| + \sup_{0 \leq x \leq t} |\hat{m}_0(t,x) - m_0(t,x)| \Big) \\ & \int_0^t \lambda_0(x) I(\{x \in [0,t] : m(t,x) < 2\delta_t\}) dx \end{aligned}$$

Since for any $t \in [0,\tau]$ we have $m(t,x) > 0$ for every $x \in [0,t)$, for each fixed x such that $m(t,x) > 0$, we have $I(\{m(t,x) < 2\delta_t\}) \to 0$ as $\delta_t \downarrow 0$. Moreover

$$0 \leq \lambda_0(x)\, I(\{m(t,x) < 2\delta_t\}) \leq \lambda_0(x),$$

and $\lambda_0(x)$ is integrable on $[0,t]$. Therefore, by the Dominated Convergence Theorem:

$$\int_0^t \lambda_0(x) I(\{x \in [0,t] : m(t,x) < 2\delta_t\}) dx \to 0 \qquad \text{as} \qquad \delta_t \downarrow 0$$

Combining the three cases we have

$$|n^{-1/2}R_{\widetilde{D}}(t)| = o_P(1) \qquad \text{for} \qquad t \in [0,\tau]$$

Therefore, for $t \in [0,\tau]$

$$n^{-1/2}\widetilde{D}(t) = n^{-1/2}\widetilde{D}_A(t) + o_P(1)$$

Part 2: Multivariate CLT

For $t \in [0, \tau]$

$$\overline{D}_A(t) = \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx \Big) \left(Z_i - \frac{m_1(t,x)}{m(t,x)} \right) = \sum_{i=1}^{n} \psi_{\overline{D},i}(t)$$

where

$$\psi_{\overline{D},i}(t) = \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x) dx \Big) \left(Z_i - \frac{m_1(t,x)}{m(t,x)} \right) \tag{45}$$

and similarly

$$\begin{aligned} \widetilde{D}_A(t) &= n(e^{\theta} - 1) \int_0^t \lambda_0(x) \Big[g(\underset{\sim}{\mathbf{m}}_{01}(t,x)) + \nabla^T g(\underset{\sim}{\mathbf{m}}_{01}(t,x)) \big(\widehat{\underset{\sim}{\mathbf{m}}}_{01}(t,x) - \underset{\sim}{\mathbf{m}}_{01}(t,x) \big) \Big] dx \\ &= n\nu(t) + \sum_{i=1}^{n} \psi_{\widetilde{D},i}(t) \end{aligned}$$

where

$$\nu(t) = (e^{\theta} - 1) \int_0^t \lambda_0(x) \frac{m_0(t,x) m_1(t,x)}{m(t,x)} dx \tag{46}$$

$$\begin{aligned} \psi_{\widetilde{D},i}(t) = (e^{\theta} - 1) \int_0^t \lambda_0(x) \Bigg[&\left(\frac{m_1(t,x)}{m(t,x)} \right)^2 \Big((1 - Z_i) Y_i(t,x) - m_0(t,x) \Big) \\ &+ \left(\frac{m_0(t,x)}{m(t,x)} \right)^2 \Big(Z_i Y_i(t,x) - m_1(t,x) \Big) \Bigg] dx \end{aligned} \tag{47}$$

At the fixed calendar times $0 < t_1 < \cdots < t_K \leq \tau$, we can write for $t_k \in \{t_1, \ldots, t_K\}$:

$$\begin{aligned} n^{-1/2} D(t_k) &= n^{-1/2} D_A(t_k) + o_P(1) \\ \implies n^{-1/2} \Big(D(t_k) - n\, \nu(t_k) \Big) &= n^{-1/2} \sum_{i=1}^{n} \Big(\psi_{\overline{D},i}(t_k) + \psi_{\widetilde{D},i}(t_k) \Big) + o_P(1) \end{aligned}$$

where $D_A(t_k) = \overline{D}_A(t_k) + \widetilde{D}_A(t_k)$. We then have the $K \times 1$ vector

$$n^{-1/2} \sum_{i=1}^{n} \begin{pmatrix} \psi_{\overline{D},i}(t_1) + \psi_{\widetilde{D},i}(t_1) \\ \vdots \\ \psi_{\overline{D},i}(t_K) + \psi_{\widetilde{D},i}(t_K) \end{pmatrix} = n^{-1/2} \sum_{i=1}^{n} \mathbf{\Psi}_i$$

where $\mathbf{\Psi}_i = \Big(\psi_{\overline{D},i}(t_1) + \psi_{\widetilde{D},i}(t_1), \ldots, \psi_{\overline{D},i}(t_K) + \psi_{\widetilde{D},i}(t_K)\Big)$ are IID $K \times 1$ vectors. Furthermore, $\mathbb{E}\mathbf{\Psi}_i = \mathbf{0}$ for all i since $\psi_{\overline{D},i}(t_k)$ are mean-zero martingales, and $\mathbb{E}\psi_{\widetilde{D},i}(t_k) = 0$ for all $i = 1, \ldots, n$, $t_k \in \{t_1, \ldots, t_K\}$, since

$$\mathbb{E}\Big[(1 - Z_i)Y_i(t, x) - m_0(t, x)\Big] = 0, \qquad \mathbb{E}\Big[Z_i Y_i(t, x) - m_1(t, x)\Big] = 0$$

Finite second moments are ensured by the assumption $\int_0^\tau \lambda_0(x)dx < \infty$: for $t \in \{t_1, \ldots, t_K\}$

$$\begin{aligned} \mathbb{E}\Big[\Big(\psi_{\overline{D},1}(t_k)\Big)^2\Big] &= \mathbb{E}\left[\int_0^{t_k} \left(Z_1 - \frac{m_1(t_k, x)}{m(t_k, x)}\right)^2 e^{\theta Z_1} \lambda_0(x) Y_1(t_k, x) dx\right] \\ &\leq (e^\theta + 1)\Lambda_0(\tau) < \infty \end{aligned}$$

using $e^{\theta Z_1} \leq (e^\theta + 1)$, $Y_1(t_k, x) \in \{0, 1\}$, and bounding the square term above by 1. Similarly,

$$|\psi_{\widetilde{D},1}(t_k)| \leq 2\,|e^\theta - 1|\,\Lambda_0(\tau) \implies \mathbb{E}\Big[\Big(\psi_{\widetilde{D},1}(t_k)\Big)^2\Big] < \infty$$

using

$$|(1 - Z_1)Y_1(t_k, x) - m_0(t_k, x)| \leq 1,\ |Z_1 Y_1(t_k, x) - m_1(t_k, x)| \leq 1,\ \left(\frac{m_z(t_k, x)}{m(t_k, x)}\right)^2 \leq 1 \text{ for } z \in \{0, 1\}$$

By the multivariate CLT for IID random vectors

$$n^{-1/2} \sum_{i=1}^{n} \begin{pmatrix} \psi_{\overline{D},i}(t_1) + \psi_{\widetilde{D},i}(t_1) \\ \vdots \\ \psi_{\overline{D},i}(t_K) + \psi_{\widetilde{D},i}(t_K) \end{pmatrix} = n^{-1/2} \sum_{i=1}^{n} \mathbf{\Psi}_i \xrightarrow{D} \mathcal{N}_K(\mathbf{0}, \Sigma)$$

where $\Sigma = \mathbb{V}(\mathbf{\Psi_1})$. By Slutsky's Lemma to account for the remainder terms:

$$n^{-1/2}\begin{pmatrix} D(t_1) - n\nu(t_1) \\ \vdots \\ D(t_K) - n\nu(t_K) \end{pmatrix} \xrightarrow{D} \mathcal{N}_K(\mathbf{0}, \Sigma)$$

Part 3: Expressions for asymptotic variances and covariances

3.1. Expression for $\mathbb{V}[\psi_{\overline{D},1}(t) + \psi_{\widetilde{D},1}(t)]$; $t \in [0, \tau]$

For $t \in [0, \tau]$,

$$\mathbb{V}[\psi_{\overline{D},1}(t) + \psi_{\widetilde{D},1}(t)] = \mathbb{V}[\psi_{\overline{D},1}(t)] + \mathbb{V}[\psi_{\widetilde{D},1}(t)] + 2\mathbb{C}\mathrm{o}\mathbb{v}[\psi_{\overline{D},1}(t)\,,\, \psi_{\widetilde{D},1}(t)] \tag{48}$$

We individually consider each of the three terms in (48).

3.1.1. Expression for $\mathbb{V}[\psi_{\overline{D},1}(t)]$; $t \in [0, \tau]$:

For $t \in [0, \tau]$ we have

$$\mathbb{V}[\psi_{\overline{D},1}(t)] = \mathbb{V}\left[\int_0^t \Big(N_1(t, dx) - e^{\theta Z_1}\lambda_0(x)Y_1(t, x)dx\Big)\left(Z_1 - \frac{m_1(t, x)}{m(t, x)}\right)\right]$$

The concerned stochastic integral being a zero-mean, locally square integrable martingale, its variance equals the expectation of its predictable quadratic variation, and we can write

$$\mathbb{V}[\psi_{\overline{D},1}(t)] = \mathbb{E}\Big[\Big(\psi_{\overline{D},1}(t)\Big)^2\Big] = \mathbb{E}\left[\int_0^t \left(Z_1 - \frac{m_1(t, x)}{m(t, x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t, x)dx\right]$$

Using $e^{\theta Z_1} = Z_1 e^{\theta} + (1 - Z_1)$ we have:

$$\mathbb{V}[\psi_{\overline{D},1}(t)] = \int_0^t \lambda_0(x)\left[e^{\theta} m_1(t, x)\left(\frac{m_0(t, x)}{m(t, x)}\right)^2 + m_0(t, x)\left(\frac{m_1(t, x)}{m(t, x)}\right)^2\right]dx \tag{49}$$

3.1.2. Expression for $\mathbb{V}[\psi_{\widetilde{D},1}(t)]$; $t \in [0, \tau]$:

For $t \in [0, \tau]$, from (47) we have

$$\psi_{\widetilde{D},1}(t) = (e^{\theta} - 1) \int_0^t \lambda_0(x) \zeta(t, x)\, dx$$

where

$$\zeta(t, x) = \left(\frac{m_1(t, x)}{m(t, x)}\right)^2 \Big((1 - Z_1)Y_1(t, x) - m_0(t, x)\Big) + \left(\frac{m_0(t, x)}{m(t, x)}\right)^2 \Big(Z_1 Y_1(t, x) - m_1(t, x)\Big)$$

Then

$$\mathbb{V}[\psi_{\widetilde{D},1}(t)] = (e^{\theta} - 1)^2 \mathbb{E}\left(\int_0^t \lambda_0(x) \zeta(t, x)\, dx\right)^2$$

since $\mathbb{E}[\psi_{\widetilde{D},1}(t)] = 0$ using $E[(1 - Z_1)Y_1(t, x) - m_0(t, x)] = 0$ and $\mathbb{E}[Z_1 Y_1(t, x) - m_1(t, x)] = 0$. Then

$$\begin{aligned}
\mathbb{V}[\psi_{\widetilde{D},1}(t)] &= (e^{\theta} - 1)^2 \mathbb{E}\left[\int_0^t \int_0^t \lambda_0(x)\, \lambda_0(y)\, \zeta(t, x)\, \zeta(t, y)\, dx\, dy\right] \\
&= (e^{\theta} - 1)^2 \int_0^t \int_0^t \lambda_0(x)\, \lambda_0(y)\, \mathbb{E}\Big[\zeta(t, x)\, \zeta(t, y)\Big]\, dx\, dy \\
&= 2\, (e^{\theta} - 1)^2 \int_0^t \int_0^y \lambda_0(x)\, \lambda_0(y)\, \mathbb{E}\Big[\zeta(t, x)\, \zeta(t, y)\Big]\, dx\, dy \qquad (50)
\end{aligned}$$

Using $Z_1^2 = Z_1$, $(1 - Z_1)^2 = 1 - Z_1$, $Z_1(1 - Z_1) = 0$ and $Y_1(t, x)Y_1(t, y) = Y_1(t, y)$ since $0 \le x \le y \le t$, we can write

$$\begin{aligned}
\mathbb{E}\Big[\zeta(t, x)\, \zeta(t, y)\Big] &= \pi^2(t, x)\pi^2(t, y)\Big(m_0(t, y) - m_0(t, x)m_0(t, y)\Big) \\
&+ (1 - \pi(t, x))^2 (1 - \pi(t, y))^2 \Big(m_1(t, y) - m_1(t, x)m_1(t, y)\Big) \\
&- \pi^2(t, x)\, (1 - \pi(t, y))^2\, m_0(t, x)\, m_1(t, y) \\
&- (1 - \pi(t, x))^2\, \pi^2(t, y)\, m_1(t, x)\, m_0(t, y) \qquad (51)
\end{aligned}$$

where $\pi(t, x) = m_1(t, x)/m(t, x)$. Substituting (51) in (50) provides the required expression for $\mathbb{V}[\psi_{\widetilde{D},1}(t)]$.

3.1.3. Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t), \psi_{\widetilde{D},1}(t)]$; $t \in [0,\tau]$:

For $t \in [0,\tau]$, we have

$$
\begin{aligned}
&\mathbb{C}\mathrm{ov}[\,\psi_{\overline{D},1}(t)\,,\,\psi_{\widetilde{D},1}(t)] \\
&= \mathbb{C}\mathrm{ov}\left[\,\psi_{\overline{D},1}(t)\,,\,(e^{\theta}-1)\int_0^t \lambda_0(y)\zeta(t,y)dy\right] \\
&= \mathbb{C}\mathrm{ov}\left[\,\psi_{\overline{D},1}(t)\,,\,(e^{\theta}-1)\int_0^t \lambda_0(y)\Big[(\pi(t,y))^2\,(1-Z_1)Y_1(t,y)+(1-\pi(t,y))^2\,Z_1Y_1(t,y)\Big]\,dy\right] \\
&\qquad - \mathbb{C}\mathrm{ov}\left[\,\psi_{\overline{D},1}(t)\,,\,(e^{\theta}-1)\int_0^t \lambda_0(y)\Big[(\pi(t,y))^2\,m_0(t,y)+(1-\pi(t,y))^2\,m_1(t,y)\Big]\,dy\right]
\end{aligned}
$$

where the second term is 0 because $\mathbb{E}[\psi_{\overline{D},1}(t)] = 0$. Therefore, for $t \in [0,\tau]$

$$
\begin{aligned}
&\mathbb{C}\mathrm{ov}[\,\psi_{\overline{D},1}(t)\,,\,\psi_{\widetilde{D},1}(t)] \\
&= \mathbb{C}\mathrm{ov}\left[\,\psi_{\overline{D},1}(t)\,,\,(e^{\theta}-1)\int_0^t \lambda_0(y)\Big[(\pi(t,y))^2\,(1-Z_1)Y_1(t,y)+(1-\pi(t,y))^2\,Z_1Y_1(t,y)\Big]\,dy\right] \\
&= (e^{\theta}-1)(\mathcal{I}_1-\mathcal{I}_2)
\end{aligned}
$$

where

$$
\begin{aligned}
\mathcal{I}_1 = \mathbb{E}\Bigg[&\left(\int_0^t N_1(t,dx)(Z_1-\pi(t,x))\right) \\
&\qquad\left(\int_0^t \lambda_0(y)\Big[(\pi(t,y))^2\,(1-Z_1)Y_1(t,y)+(1-\pi(t,y))^2\,Z_1Y_1(t,y)\Big]\,dy\right)\Bigg] \\
\mathcal{I}_2 = \mathbb{E}\Bigg[&\left(\int_0^t e^{\theta Z_1}\lambda_0(x)Y_1(t,x)(Z_1-\pi(t,x))dx\right) \\
&\qquad\left(\int_0^t \lambda_0(y)\Big[(\pi(t,y))^2\,(1-Z_1)Y_1(t,y)+(1-\pi(t,y))^2\,Z_1Y_1(t,y)\Big]\,dy\right)\Bigg]
\end{aligned}
$$

We consider $\mathcal{I}_2$ first:

$$
\begin{aligned}
\mathcal{I}_2 = \mathbb{E}&\left[\left(\int_0^t e^{\theta}\lambda_0(x)Z_1Y_1(t,x)(Z_1-\pi(t,x))dx\right)\left(\int_0^t \lambda_0(y)(1-\pi(t,y))^2Z_1Y_1(t,y)dy\right)\right] \\
+\,\mathbb{E}&\left[\left(\int_0^t \lambda_0(x)(1-Z_1)Y_1(t,x)(Z_1-\pi(t,x))dx\right)\left(\int_0^t \lambda_0(y)(\pi(t,y))^2(1-Z_1)Y_1(t,y)dy\right)\right]
\end{aligned}
$$

using $e^{\theta Z_1} = Z_1 e^{\theta} + (1 - Z_1)$ along with $Z_1(1 - Z_1) = 0$. Since $Z_1^2 = Z_1$, $(1 - Z_1)^2 = (1 - Z_1)$, and $Y_1(t, x) = I(X_1(t) \geq x)$, we can write

$$\begin{aligned}
\mathcal{I}_2 = e^{\theta}\mathbb{E}\Bigg[Z_1\Bigg(\int_0^{X_1(t)} \lambda_0(x)(1 - \pi(t, x))dx\Bigg)\Bigg(\int_0^{X_1(t)} \lambda_0(y)(1 - \pi(t, y))^2 dy\Bigg)\Bigg] \\
+ \mathbb{E}\Bigg[(1 - Z_1)\Bigg(\int_0^{X_1(t)} \lambda_0(x)(-\pi(t, x))dx\Bigg)\Bigg(\int_0^{X_1(t)} \lambda_0(y)(\pi(t, y))^2 dy\Bigg)\Bigg] \\
= e^{\theta}\, p\, \mathbb{E}[A_1(X_1(t))B_1(X_1(t))|Z_1 = 1] + \, (1 - p)\, \mathbb{E}[A_0(X_1(t))B_0(X_1(t))|Z_1 = 0]
\end{aligned}$$

where

$$A_z(u) = \int_0^u \lambda_0(x)(z - \pi(t, x))dx \qquad \text{and} \qquad B_z(u) = \int_0^u \lambda_0(x)(z - \pi(t, x))^2 dx$$

Since

$$A_z(0) = 0 \qquad \text{and} \qquad B_z(0) = 0 \qquad \text{for} \qquad z \in \{0, 1\}$$

we can write for $z \in \{0, 1\}$

$$A_z(X_1(t))\, B_z(X_1(t)) = \int_0^t \Big[A_z^{'}(x)B_z(x) + A_z(x)B_z^{'}(x)\Big]\, I(x \leq X_1(t))\, dx$$

and by taking the conditional expectation on both sides

$$\mathbb{E}[A_z(X_1(t))B_z(X_1(t))|Z_1 = z] = \int_0^t \Big[A_z^{'}(x)B_z(x) + A_z(x)B_z^{'}(x)\Big] P(X_1(t) \geq x|Z_1 = z)\, dx$$

Therefore,

$$\begin{aligned}
\mathcal{I}_2 = e^{\theta}\, p \int_0^t \Big[A_1^{'}(x)B_1(x) + A_1(x)B_1^{'}(x)\Big] P[X_1(t) \geq x|Z_1 = 1]\, dx \\
+ (1 - p) \int_0^t \Big[A_0^{'}(x)B_0(x) + A_0(x)B_0^{'}(x)\Big] P[X_1(t) \geq x|Z_1 = 0]\, dx
\end{aligned}$$

We can similarly show that

$$\mathcal{I}_1 = e^{\theta} p \int_0^t A_1^{'}(x) B_1(x) P[X_1(t) \geq x | Z_1 = 1]\, dx + (1-p) \int_0^t A_0^{'}(x) B_0(x) P[X_1(t) \geq x | Z_1 = 0]\, dx$$

Putting them back together, we get

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\,\psi_{\overline{D},1}(t)\,,\ \psi_{\widetilde{D},1}(t)\,] = (e^{\theta}-1)(\mathcal{I}_1 - \mathcal{I}_2) \\
&= -(e^{\theta}-1)\Bigg[e^{\theta} p \int_0^t A_1(x) B_1^{'}(x) P[X_1(t) \geq x | Z_1 = 1]\, dx \\
&\qquad\qquad + (1-p) \int_0^t A_0(x) B_0^{'}(x) P[X_1(t) \geq x | Z_1 = 0]\, dx \Bigg] \\
&= -(e^{\theta}-1)\Bigg[e^{\theta} p \int_0^t \left(\int_0^x \lambda_0(y)(1-\pi(t,y))\, dy \right) \lambda_0(x)(1-\pi(t,x))^2\, F_{U_1}(t-x) \overline{F}_{V_1|Z_1=1}(x) dx \\
&\qquad + (1-p) \int_0^t \left(\int_0^x \lambda_0(y)(-\pi(t,y))\, dy \right) \lambda_0(x)(\pi(t,x))^2\, F_{U_1}(t-x) \overline{F}_{V_1|Z_1=0}(x) dx \Bigg]
\end{aligned} \tag{52}$$

using

$$\begin{aligned}
P[X_1(t) \geq x | Z_1 = z] &= P[V_1 \geq x\,, t - U_1 \geq x | Z_1 = z] \\
&= P[U_1 \leq t - x] P[V_1 \geq x | Z_1 = z] \\
&= F_{U_1}(t-x) \overline{F}_{V_1|Z_1=z}(x)
\end{aligned}$$

since U_1 is assumed to be independent of (V_1, Z_1). Substituting expressions (49), (50), and (52) in (48) provides the desired asymptotic variance expression.

3.2. Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1) + \psi_{\widetilde{D},1}(t_1)\,, \psi_{\overline{D},1}(t_2) + \psi_{\widetilde{D},1}(t_2)]$; $0 \leq t_1 < t_2 \leq \tau$

For $0 \leq t_1 < t_2 \leq \tau$:

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1) + \psi_{\widetilde{D},1}(t_1)\,, \psi_{\overline{D},1}(t_2) + \psi_{\widetilde{D},1}(t_2)] \\
&= \mathbb{V}[\psi_{\overline{D},1}(t_1)] + \mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\ \psi_{\widetilde{D},1}(t_2)] + \mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_2)\,,\ \psi_{\widetilde{D},1}(t_1)] + \mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\ \psi_{\widetilde{D},1}(t_2)]
\end{aligned} \tag{53}$$

where

$$\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1), \psi_{\overline{D},1}(t_2)] = \mathbb{V}[\psi_{\overline{D},1}(t_1)]$$

using the independent increments property of the zero-mean, square integrable martingale $\psi_{\overline{D},1}(t)$.

3.2.1 Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\,\psi_{\tilde{D},1}(t_2)]$

For $0 \leq t_1 < t_2 \leq \tau$

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\,\psi_{\tilde{D},1}(t_2)] = \mathbb{E}[\psi_{\overline{D},1}(t_1)\,\psi_{\tilde{D},1}(t_2)] \\
&= (e^{\theta}-1)\mathbb{E}\left[\psi_{\overline{D},1}(t_1)\left(\int_0^{t_2} \lambda_0(x)(1-\pi(t_2,x))^2 Z_1 Y_1(t_2,x)dx\right)\right] \\
&\qquad + (e^{\theta}-1)\mathbb{E}\left[\psi_{\overline{D},1}(t_1)\left(\int_0^{t_2} \lambda_0(x)(\pi(t_2,x))^2 (1-Z_1) Y_1(t_2,x)dx\right)\right] \\
&= (e^{\theta}-1)(\mathcal{I}_1+\mathcal{I}_2) \qquad (54)
\end{aligned}$$

Note that

$$\begin{aligned}
\pi(t,x) = \frac{m_1(t,x)}{m(t,x)} = \frac{\mathbb{E}[Z_1Y_1(t,x)]}{\mathbb{E}[Y_1(t,x)]} &= \frac{pF_{U_1}(t-x)\overline{F}_{V_1|Z_1=1}(x)}{pF_{U_1}(t-x)\overline{F}_{V_1|Z_1=1}(x)+(1-p)F_{U_1}(t-x)\overline{F}_{V_1|Z_1=0}(x)} \\
&= \frac{p\overline{F}_{V_1|Z_1=1}(x)}{p\overline{F}_{V_1|Z_1=1}(x)+(1-p)\overline{F}_{V_1|Z_1=0}(x)}
\end{aligned}$$

where the third equality hold since U_1 is independent of (V_1, Z_1). The final expression for $\pi(t,x)$ being independent of t, we drop the t and use $\pi(x)$ instead. We first consider $\mathcal{I}_1$:

$$\begin{aligned}
\mathcal{I}_1 &= \mathbb{E}\left[\left(\int_0^{t_2} \lambda_0(x)(1-\pi(x))^2 Z_1Y_1(t_2,x)dx\right)\psi_{\overline{D},1}(t_1)\right] \\
&= \mathbb{E}\left[\left(\int_0^{t_2} \lambda_0(x)(1-\pi(x))^2 Z_1Y_1(t_2,x)dx\right)\left(\int_0^{t_1}\Big(N_1(t_1,dy)-e^{\theta Z_1}\lambda_0(y)Y_1(t_1,y)dy\Big)\Big(Z_1-\pi(y)\Big)\right)\right] \\
&= \int_0^{t_2} \lambda_0(x)(1-\pi(x))^2\,\mathbb{E}\left[Z_1Y_1(t_2,x)\int_0^{t_1} N_1(t_1,dy)\,(1-\pi(y))\right]dx \\
&\qquad - \int_0^{t_2} \lambda_0(x)(1-\pi(x))^2\,\mathbb{E}\left[e^{\theta}Z_1Y_1(t_2,x)\int_0^{t_1} \lambda_0(y)Y_1(t_1,y)(1-\pi(y))dy\right]dx
\end{aligned}$$

where the last equality follows by using $Z_1(1-Z_1)=0$. By conditioning on $Z_1=1$, using

$$Y_1(t_2,x)\,Y_1(t_1,y) = I(U_1 \le (t_2-x)\wedge(t_1-y))\,I(V_1 \ge x\vee y)$$
$$Y_1(t_2,x)N_1(t_1,dy) = 0 \qquad \text{if} \qquad y<x$$

and exchanging the integration and expectation using Fubini, we can write

$$\begin{aligned}
\mathcal{I}_1 &= p\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_x^{t_1}(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,f_{V_1|Z_1=1}(y)\,dy\right]dx\\
&- p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_0^{t_1}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(x\vee y)dy\right]dx\\
&= \mathcal{I}_{11}-\mathcal{I}_{12}
\end{aligned}$$

Writing $f_{V_1|Z_1=1}(y) = e^{\theta}\lambda_0(y)\overline{F}_{V_1|Z_1=1}(y)$ in $\mathcal{I}_{11}$, and splitting up the inner integral in $\mathcal{I}_{12}$, we can write

$$\begin{aligned}
\mathcal{I}_1 &= p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_x^{t_1}\lambda_0(y)(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(y)\,dy\right]dx\\
&- p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_0^{x}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(x)dy\right]dx\\
&- p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_x^{t_1}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(y)dy\right]dx\\
&= -\,p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\,\overline{F}_{V_1|Z_1=1}(x)\left[\int_0^{x\wedge t_1}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,dy\right]dx
\end{aligned}$$

Similarly, one can show:

$$\mathcal{I}_2 = -(1-p)\int_0^{t_2}\lambda_0(x)\pi^2(x)\,\overline{F}_{V_1|Z_1=0}(x)\left[\int_0^{x\wedge t_1}\lambda_0(y)(-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,dy\right]dx$$

Substituting these expressions in (54)

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)] \\
&= -(e^{\theta}-1)\sum_{z=0}^{1} P(Z_1=z)\,e^{\theta z}\int_0^{t_2}\lambda_0(x)\,(z-\pi(x))^2\,\overline{F}_{V_1|Z_1=z}(x) \\
&\qquad\qquad \left[\int_0^{x\wedge t_1}\lambda_0(y)\,(z-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,dy\right]dx
\end{aligned} \tag{55}$$

3.2.2 Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_2)\,,\,\psi_{\widetilde{D},1}(t_1)]$

Using the same steps as in **3.2.1**, one can show for $0 \le t_1 < t_2 \le \tau$

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_2)\,,\,\psi_{\widetilde{D},1}(t_1)] \\
&= -(e^{\theta}-1)\sum_{z=0}^{1} P(Z_1=z)\,e^{\theta z}\int_0^{t_1}\lambda_0(x)\,(z-\pi(x))^2 \\
&\qquad\qquad F_{U_1}(t_1-x)\,\overline{F}_{V_1|Z_1=z}(x)\left(\int_0^{x}\lambda_0(y)\,(z-\pi(y))\,dy\right)dx
\end{aligned} \tag{56}$$

3.2.3 Expression for $\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)]$ For $0 \le t_1 < t_2 \le \tau$

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)] = \mathbb{E}[\psi_{\widetilde{D},1}(t_1)\;\psi_{\widetilde{D},1}(t_2)] \\
&= (e^{\theta}-1)^2\mathbb{E}\left[\left(\int_0^{t_1}\lambda_0(x)\zeta(t_1,x)dx\right)\left(\int_0^{t_2}\lambda_0(y)\zeta(t_2,y)dy\right)\right]
\end{aligned}$$

Now,

$$\begin{aligned}
&\mathbb{E}\left[\left(\int_0^{t_1}\lambda_0(x)\zeta(t_1,x)dx\right)\left(\int_0^{t_2}\lambda_0(y)\zeta(t_2,y)dy\right)\right] \\
&= \int_0^{t_1}\int_0^{t_2}\lambda_0(x)\lambda_0(y)\Bigg[(1-\pi(x))^2\,(1-\pi(y))^2\mathbb{C}\mathrm{ov}\Big(Z_1Y_1(t_1,x)\,,\,Z_1Y_1(t_2,y)\Big) \\
&\qquad + (1-\pi(x))^2\,\pi^2(y)\mathbb{C}\mathrm{ov}\Big(Z_1Y_1(t_1,x)\,,\,(1-Z_1)Y_1(t_2,y)\Big) \\
&\qquad + \pi^2(x)\,(1-\pi(y))^2\mathbb{C}\mathrm{ov}\Big((1-Z_1)Y_1(t_1,x)\,,\,Z_1Y_1(t_2,y)\Big) \\
&\qquad\qquad + \pi^2(x)\,\pi^2(y)\mathbb{C}\mathrm{ov}\Big((1-Z_1)Y_1(t_1,x)\,,\,(1-Z_1)Y_1(t_2,y)\Big)\Bigg]\,dy\,dx
\end{aligned}$$

Using $Z_1(1-Z_1)=0$ we can write

$$
\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)]\\
&=(e^{\theta}-1)^2\int_0^{t_1}\int_0^{t_2}\lambda_0(x)\lambda_0(y)\\
&\qquad\Big[(1-\pi(x))^2\,(1-\pi(y))^2\Big(\mathbb{E}[Z_1Y_1(t_1,x)Y_1(t_2,y)]-m_1(t_1,x)m_1(t_2,y)\Big)\\
&\qquad+(1-\pi(x))^2\,\pi^2(y)\Big(-m_1(t_1,x)m_0(t_2,y)\Big)+\pi^2(x)(1-\pi(y))^2\Big(-m_0(t_1,x)m_1(t_2,y)\Big)\\
&\qquad\quad+\pi^2(x)\pi^2(y)\Big(\mathbb{E}[(1-Z_1)Y_1(t_1,x)Y_1(t_2,y)]-m_0(t_1,x)m_0(t_2,y)\Big)\Big]\,dy\,dx
\end{aligned}
\tag{57}
$$

where

$$
\begin{aligned}
\mathbb{E}[Z_1Y_1(t_1,x)Y_1(t_2,y)]&=p\,F_{U_1}((t_1-x)\wedge(t_2-y))\,\overline{F}_{V_1|Z_1=1}(x\vee y)\\
\mathbb{E}[(1-Z_1)Y_1(t_1,x)Y_1(t_2,y)]&=(1-p)\,F_{U_1}((t_1-x)\wedge(t_2-y))\,\overline{F}_{V_1|Z_1=0}(x\vee y)
\end{aligned}
$$

Finally, substituting the expressions (49), (55), (56), and (57) in (53), we obtain the required expressions for the asymptotic covariance.

Proof of Theorem 2

For $t\in[0,\tau]$ and for fixed number n of subjects entering the trial over the course of the accrual period:

$$
\widetilde{D}(t)=(e^{\theta}-1)\sum_{i=1}^{n}\int_0^t\lambda_0(x)Z_iY_i(t,x)\left(Z_i-\frac{\sum_{j\in R(t,x)}Z_j}{n(t,x)}\right)dx
$$

using $e^{\theta Z_i}=Z_ie^{\theta}+(1-Z_i)$ and $n(t,x)=\sum_{i=1}^{n}Y_i(t,x)$. Since the summation over $j\in R(t,x)$ includes the ith subject as well when $Y_i(t,x)=1$, we can write

$$
\widetilde{D}(t)=(e^{\theta}-1)\sum_{i=1}^{n}\int_0^t\lambda_0(x)Z_iY_i(t,x)\left(1-\frac{1+\sum_{j\in R(t,x)\setminus\{i\}}Z_j}{1+|R(t,x)\setminus\{i\}|}\right)dx
$$

Conditioning on $B_i = \{j = 1, \ldots, n : j \neq i, j \in R(t,x)\}$ along with Z_i, and $\{X_i(t) \geq x\}$:

$$\mathbb{E}\widetilde{D}(t) = (e^\theta - 1) \sum_{i=1}^{n} \int_0^t \lambda_0(x) \mathbb{E}\left[Z_i Y_i(t,x) \left(1 - \frac{1 + \mathbb{E}[\sum_{j \in B_i} Z_j | B_i, Z_i, X_i(t) \geq x]}{1 + |B_i|} \right) \right] dx$$

Since Z_j for $j \neq i$ are conditionally distributed as Bernoulli(p), conditionally

$$\sum_{j \in B_i} Z_j \sim \text{binomial}(|B_i|, \pi(t,x)) \qquad \text{where} \qquad \pi(t,x) = P[Z_j = 1 | X_j(t) \geq x]$$

which gives us

$$\mathbb{E}\widetilde{D}(t) = n\,(e^\theta - 1) \int_0^t \lambda_0(x) \mathbb{E}\left[Z_1 Y_1(t,x)(1 - \pi(t,x)) \frac{|B_1|}{1 + |B_1|} \right] dx$$

Conditioned on $Z_1 = 1$ and $\{X_1(t) \geq x\}$, $|B_1| \sim \text{binomial}(n-1, q(t,x))$ where $q(t,x) = P[X_1(t) \geq x | Z_1 = 1]$. Then

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{1 + |B_1|} \middle| Z_1 = 1, X_1(t) \geq x \right] &= \sum_{j=0}^{n-1} \frac{1}{j+1} \binom{n-1}{j} q(t,x)^j (1 - q(t,x))^{n-1-j} \\
&= \frac{1}{n\,q(t,x)} \sum_{j=1}^{n} \binom{n}{j} q(t,x)^j (1 - q(t,x))^{n-j} \\
&= \frac{1 - (1 - q(t,x))^n}{n\,q(t,x)}
\end{aligned}$$

Plugging this back into $\mathbb{E}\widetilde{D}(t)$ and using $\mathbb{E}[Z_1 Y_1(t,x)] = p\,q(t,x)$, we have

$$\mathbb{E}\widetilde{D}(t) = n\,p\,(e^\theta - 1) \int_0^t \lambda_0(x)(1 - \pi(t,x)) q(t,x) \left[1 - \frac{1 - (1 - q(t,x))^n}{n\,q(t,x)} \right] dx$$

Noting that $\pi(t,x) = P[Z_1 = 1 | X_1(t) \geq x] = m_1(t,x)/m(t,x)$, and $q(t,x) = P[X_1(t) \geq x | Z_1 = 1] = m_1(t,x)/p$, we can show $n^{-1/2}(\mathbb{E}\widetilde{D}(t) - n\nu(t)) = o(1)$ for all $t \in [0, \tau]$.

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